

Solving Systems Algebraically by Substitution or Elimination

Answer Key

Grade 8 / Pre-Algebra Expressions and Equations

Quick Answers

1. $x = 10, y = 30$

2. $x = 11, y = 7$

3. $x = 15, y = 3$

4. $x = 2, y = 3$

5. $x = 2, y = 5$

6. $x = 90, y = 110$

7. $x = 10, y = 70$

8. $x = 6, y = 2$

9. —

10. $x = 7, y = 5$

Curriculum

Level: Grade 8 / Pre-Algebra Expressions and Equations

8.EE.C.8 (sub-codes 8.EE.C.8a–c for graphical/algebraic/word-problem emphasis) – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–5 of 5 across 11 tagged checks.

1. School play. $x + y = 40, y = 3x$.

Step 1. The second equation already gives y by itself, so substitute $3x$ for y in the first: $x + 3x = 40$, that is $4x = 40$.

Step 2. So $x = 10$ adult tickets, and $y = 3(10)$ child tickets.

$x = 10, y = 30$

Check both equations: $10 + 30 = 40$ tickets in total, and 30 is indeed 3 times 10.

2. Garden. $2x + 2y = 36, x - y = 4$.

Step 1. The second equation rearranges in one step to $x = y + 4$, which is what makes substitution the efficient choice here.

Step 2. Substituting: $2(y + 4) + 2y = 36$, so $4y + 8 = 36$ and $4y = 28$.

Step 3. Then $y = 7$ for the width, and $x = 7 + 4$ for the length, both in metres.

$x = 11, y = 7$

Check: the perimeter is $2(11) + 2(7) = 36$ metres, and the length exceeds the width by 4 metres.

3. Boat and current. $x + y = 18, x - y = 12$.

Step 1. The y terms are $+y$ and $-y$ — already opposites — so adding the equations removes y with no scaling at all.

Step 2. Adding: $2x = 30$, so the boat's speed in still water is $x = 15$ km/h. Substituting back into $x + y = 18$ gives the current.

$x = 15, y = 3$

Check: downstream the speeds add, $15 + 3 = 18$; upstream they subtract, $15 - 3 = 12$. Both readings are reproduced, so the current runs at 3 km/h.

4. Cafe order. $4x + 3y = 17$, $4x + 5y = 23$.

Step 1. The x coefficients are identical, so *subtracting* the first equation from the second removes x . Adding would give $8x + 8y$, which removes nothing.

Step 2. Subtracting: $(4x + 5y) - (4x + 3y) = 23 - 17$, so $2y = 6$ and a juice costs $y = 3$ dollars.

Step 3. Back into the first equation: $4x + 9 = 17$, so $4x = 8$.

$$x = 2, y = 3$$

Check: $4(2) + 3(3) = 17$ and $4(2) + 5(3) = 23$. Same coefficient means subtract; opposite coefficients mean add.

5. Stationery. $3x + 2y = 16$, $5x + 4y = 30$.

Step 1. Compare the y terms: 2 and 4. Doubling the first equation turns $2y$ into $4y$, so only one equation needs scaling.

Step 2. $2(3x + 2y = 16)$ gives $6x + 4y = 32$. Subtracting the second equation: $(6x + 4y) - (5x + 4y) = 32 - 30$, so $x = 2$ dollars for a pen.

Step 3. Then $3(2) + 2y = 16$, so $2y = 10$.

$$x = 2, y = 5$$

Check: $3(2) + 2(5) = 16$ and $5(2) + 4(5) = 30$. Scaling an equation never changes its solutions — both sides are multiplied by the same number.

6. Ticket sales. $x + y = 200$, $5x + 3y = 780$.

Step 1. Multiplying the count equation by 3 matches the $3y$ term: $3x + 3y = 600$.

Step 2. Subtracting from the money equation: $(5x + 3y) - (3x + 3y) = 780 - 600$, so $2x = 180$ and $x = 90$ adult tickets.

Step 3. Then $90 + y = 200$.

$$x = 90, y = 110$$

Check: $90 + 110 = 200$ tickets, and $5(90) + 3(110) = 450 + 330 = 780$ dollars. Multiplying by 5 instead of 3 works equally well and eliminates x .

7. Two gyms. $y = 20 + 5x$, $y = 50 + 2x$.

Step 1. Both equations give y on its own, so the two right-hand sides must be equal at the moment the plans cost the same: $20 + 5x = 50 + 2x$.

Step 2. Collecting terms: $3x = 30$, so $x = 10$ months.

Step 3. Either equation gives the cost: $y = 20 + 5(10)$, and the other agrees, $50 + 2(10)$.

$$x = 10, y = 70$$

What the answer means: at 10 months both gyms have cost 70 dollars. Before 10 months Gym A is cheaper because of its smaller joining fee; after 10 months Gym B is cheaper because it charges less each month.

8. Crates. $4x + 3y = 30$, $3x + 5y = 28$.

Step 1. Neither variable matches, so pick one and build a common multiple. For x , the coefficients 4 and 3 share the multiple 12: scale the first equation by 3 and the second by 4.

Step 2. $12x + 9y = 90$ and $12x + 20y = 112$. Subtracting the first from the second: $11y = 22$, so a pear crate weighs $y = 2$ kilograms.

Step 3. Back into $4x + 3(2) = 30$: $4x = 24$.

$$x = 6, y = 2$$

Check: $4(6) + 3(2) = 30$ and $3(6) + 5(2) = 28$. Eliminating y instead — scale by 5 and 3 to reach $15y$ — gives the same pair, and is worth trying as a second route.

9. Two receipts. $2x + 3y = 19$, $4x + 6y = 34$.

Step 1. Doubling the first equation gives $4x + 6y = 38$. The second receipt says the same left-hand side equals 34.

Step 2. Subtracting the two: $0 = 4$. Every variable is gone and what remains is false, so no pair (x, y) can satisfy both equations. The system has **no solution**, and the two receipts cannot both be right.

Step 3. Geometrically: both equations have the same slope — doubling one gives the same left side — so the lines are parallel and distinct. Parallel lines never meet, and a solution is exactly a meeting point.

What to look for in the student's answer: the sentence “ $0 = 4$, which is impossible” is the whole insight. A student who writes x and y values here has made an arithmetic slip; a student who writes “no solution” and says why the lines are parallel has understood the case. Note the contrast worth teaching beside it: if the second receipt had said 38 dollars, the equations would be the same line, and every price pair on it would fit — infinitely many solutions rather than none.

10. Challenge: sports equipment. $3x + 2y = 31$, $5x + 4y = 55$.

Step 1. Part (a): the y coefficients are 2 and 4, so double the first equation: $6x + 4y = 62$.

Step 2. Subtracting the second equation: $(6x + 4y) - (5x + 4y) = 62 - 55$, so $x = 7$ dollars for a football.

Step 3. Then $3(7) + 2y = 31$, so $2y = 10$.

$x = 7, y = 5$

Step 4. Check both purchases: $3(7) + 2(5) = 31$ and $5(7) + 4(5) = 55$. Both hold, so the prices are correct.

Step 5. Part (b) — open response, read the reasoning. At the prices just found, 6 footballs and 4 basketballs cost $6(7) + 4(5) = 62$ dollars, not the 60 the catalogue claims. So the third equation $6x + 4y = 60$ is not satisfied by the only pair that satisfies the first two.

What a full answer says: two lines in the plane usually meet at one point, and that point is the only candidate for a solution of the whole set. A third line either passes through that point or it does not; here it does not, so the three equations have no common solution even though any two of them are consistent. Accept any argument that identifies the unique intersection of the first two equations and tests the third against it — by substitution, or by noting that $6x + 4y$ is exactly twice $3x + 2y$ and so must equal 62.